

Sumudu Ishadi Ratnayake

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lilvamp237.github.io/sumudu-ratnayake-portfolio

PROFESSIONAL SUMMARY

Final-year Computer Science undergraduate and current Software Engineer Intern at WSO2, with a strong foundation in scalable system architecture and Agentic AI frameworks. Skilled in full-stack web development and building AI-driven pipelines using Python, Java, React, and Next.js. Proven track record of developing award-winning applications and integrating robust multi-agent systems to deliver intelligent, reliable, and automated solutions. Actively engaged in academic research, with published frameworks bridging semantic web technologies, and inclusive digital infrastructure.

EDUCATION

General Sir John Kotelawala Defence University <i>BSc. (Hons) in Computer Science; GPA: 3.78/4.0</i> Honors: Vice Chancellor's List (2024), Dean's List (2025, 2023)	Colombo, Sri Lanka 2023 – Present
St. Lawrence's Convent <i>GCE Advanced Level; Physical Science</i>	Colombo, Sri Lanka 2021

EXPERIENCE

Intern Software Engineer - Solutions Business Unit, Engineering <i>WSO2</i> Enhancing FHIR Terminology Server using Ballerina and relational databases to achieve conformance with the HL7 FHIR Terminology Ecosystem Implementation Guide.	June 2026 – Present Colombo 04, Sri Lanka
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CERTIFICATIONS

AI & Data: Neo4j Certified Professional, ML with Python (Anaconda), Azure AI Fundamentals (Microsoft), OCI 2025 AI Foundations (Oracle).
Software: Postman API Student Expert, GitHub Copilot (Microsoft), Unity Essentials.

TECHNICAL SKILLS

Languages: Python, Java, C++, TypeScript, SQL, HTML/CSS
Frameworks: React, Next.js, TensorFlow, Scikit-Learn, LangChain
Developer Tools: Git, Neo4j, Supabase, Firebase, Docker, Unity
Concepts: OOP, Data Structures & Algorithms, Multi-Agent Systems, RAG Workflows

PROJECTS

Ontology-Driven VR Learning Platform (Final Year Project) <i>Unity, C#, Neo4j, Graph Databases</i> Developing a VR environment for DHH students using ontologies and Neo4j to map semantic relationships for interactive, accessible learning paths. github.com	2026
Agentic Guest Experience Optimizer <i>Python, Scikit-learn, Pandas, Streamlit</i> Engineered an AI-driven hospitality pipeline utilizing K-Means and Random Forest, integrating a rule-based agentic workflow to automate hyper-personalized guest communications. github.com	2025
eGOV Comprehensive Platform <i>Next.js Monorepo, Docker, Supabase, PostgreSQL</i> Co-architected a scalable multi-portal monorepo in a containerized Docker environment with optimized server-side rendering and modular governance components. github.com	2025
Market Analysis Platform <i>Agent-Based Modeling, RAG, Python</i> Engineered an agent-based platform for real-time market analysis, utilizing autonomous agents to simulate and predict market trends. github.com	2026
CareSync Healthcare Platform <i>Next.js, Supabase, PostgreSQL</i> Co-developed a full-stack shift-management application that secured 1st Runner-Up at the national Code with WIE 2025 hackathon for its scheduling efficiency. github.com	2025

RESEARCH & PUBLICATIONS

Context-Aware Assistive Technology <i>SSE Framework</i>	IEEE ICCUBEA 2026 - India
Bridging the Semantic Gap in Deaf Education <i>Ontology-Driven AI</i>	JCoS 2026 - Sri Lanka
AI-Driven Inclusive Digital Infrastructure <i>UCLF Framework</i>	ICSIDS 2025 - India
Green Cloud Computing Framework <i>UGCCOF Model</i>	18th IRC, KDU - Sri Lanka
Sleepzy Lucid Dream Induction <i>PASI Framework</i>	18th IRC, KDU - Sri Lanka
Sustainability Awareness via Mobile Apps <i>UI/UX Gap Analysis</i>	18th IRC, KDU - Sri Lanka

ACHIEVEMENTS

Champion, Data Odyssey 2024 <i>1st Place among 100+ participants</i>	2024
1st Runner-Up, InnovateX University Edition 2025 <i>National Hackathon Challenge by Zebra Technologies</i>	2025
1st Runner-Up, Code with WIE 2025 <i>Organized by IEEE WIE Sri Lanka Section</i>	2025
National Finalist <i>Tech-Triathlon by Rootcode (Top 8), UoC ICDS Mini Hackathon, CodeRally 6.0</i>	2025

SOFT SKILLS

Analytical problem-solver, collaborative team player, and adaptive learner with strong technical communication skills.

LEADERSHIP & EXTRA-CURRICULAR

Secretary <i>BCS Student Chapter at KDU</i>	2025 – 2026
Head of Industry Engagement <i>IEEE Computer Society KDU</i>	2025 – 2026

REFERENCES

Dr. Kaneeka Vidanage <i>Senior Lecturer (Gr. II); vidanage_bvki@kdu.ac.lk</i>	Head of Department (Computer Science), KDU <i>+94 71 354 3786</i>
Mrs. Geethanjali Jayasuriya <i>geetha.jayasuriya@courts.gov.vic.au</i>	Senior Courtlink Developer, CSV Digital - IT <i>+61 41 914 1193</i>

I hereby confirm that the above information is true and correct.